

# Strengthening steel bridge sections using CFRP laminates

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Received 17 December 2000; accepted 15 February 2001

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## Abstract

This paper presents results from an experimental investigation to determine the feasibility of using carbon fiber reinforced polymer (CFRP) epoxy laminates to repair steel composite bridge members. Six specimens, each consisting of a 6.1 m long W8 × 24 wide flange A36 steel beam acting compositely with a 0.114 m thick by 0.71 m wide reinforced concrete slab, were first loaded past yield of the tension flange to simulate severe service distress. The damaged specimens were then repaired using 3.65 m lengths of 2 or 5 mm thick CFRP laminates bonded to the tension flange and tested to failure. The results indicated significant ultimate strength gains but more modest improvement in the elastic response. Non-linear finite element analyses were in good agreement with the experimental results. The study suggests that it is feasible to strengthen steel composite members using CFRP laminates. © 2001 Elsevier Science Ltd. All rights reserved.

**Keywords:** A. Laminates; Carbon fiber reinforced polymer

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## 1. Introduction

Fiber reinforced polymers have been successfully used to strengthen structural elements made of concrete, masonry and wood. Numerous applications have been cited for bridge structures (e.g. repair of prestressed concrete and wood bridges in Switzerland [10], multi-span hollow box girder bridge in Germany [11], slab bridges in Japan [12], and reinforced/prestressed concrete bridges in the United States [13]).

Although there have been some studies which involved strengthening steel members (e.g. Refs. [3,4]), this extension has not been so widespread since it poses a different set of problems. First, the likelihood of lateral buckling makes it necessary to fabricate composite steel sections where the compression flange is continuously supported by a reinforced concrete slab. These sections are much more difficult to fabricate and test in comparison with rectangular concrete or wood sections. Second, steel's high strength and stiffness make it a more difficult material to strengthen especially since carbon fiber reinforced polymers (CFRP) that are commonly used have a smaller modulus. This means that substantial load transfer can only take place *after* the steel

has yielded. Finally, for this system, the CFRP/adhesive bond is the weakest link and will always control the mode of failure. Thus, consideration of surface preparation alone will not suffice. Instead, more attention must be paid to augment the capacity of the adhesive through the use of appropriate fasteners.

Successful strengthening of steel bridge sections with CFRP would be particularly useful as steel stringer bridges constitute a large segment of aging bridges in the United States that are nearing the end of their useful service life. If service life could be extended, highway authorities would have access to a lightweight, speedy and therefore economical strengthening method that would minimally disrupt traffic.

The opportunity to use CFRP laminates to strengthen steel came unexpectedly. In 1992, the University of South Florida, Tampa, FL were developing a steel composite bridge model to evaluate the restraint effect of elastomeric bridge bearings [7]. For this study, the girder spacing, slab thickness or the number of shear connectors was not critically important. Consequently, advantage was taken to alter these parameters to boost the ultimate capacity of the slab and to ensure that if the concrete slab were cut longitudinally, three identical steel composite beams would result. In addition, an identical second bridge was fabricated to provide additional specimens.

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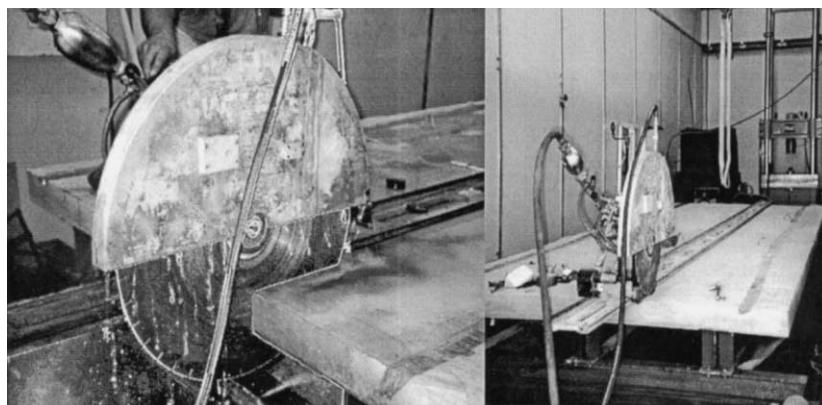


Fig. 1. Cutting bridge into three composite beam sections.

Following completion of the experimental phase of the bearing restraint study, the concrete slabs of the two bridges were cut (Fig. 1) to obtain the six test specimens used in this study. This paper presents a brief summary of the technical problems that were overcome and highlights some of the principal findings. A comprehensive report describing all aspects of the research is available [6].

## 2. Objectives

The overall goal of the study was to assess the feasibility of using CFRP laminates for strengthening steel composite bridge members. Since only distressed sections would be strengthened it was necessary to simulate such damage prior to any strengthening. The main objectives of the study may be summarized as follows:

1. To develop a procedure that could be used for strengthening steel composite members using CFRP laminates.
2. To evaluate the potential benefit of strengthening damaged steel composite specimens using CFRP laminates.
3. To assess the extent to which experimental results may be predicted using available non-linear finite element computer programs.

## 3. Specimen details

As mentioned already, the six specimens used in the study were obtained by cutting each of two concrete/steel composite bridge models into three parts (Fig. 1). The resulting cross-section is shown in Fig. 2. It consisted of a 114 mm thick, 710 mm wide reinforced concrete slab that was attached to a W8 × 24 steel section by 36 shear connectors. The spacing of the 50 mm long, 19 mm diameter stud-type shear connectors was varied — they were more closely spaced near the supports than at mid-span. The concrete slab deck was reinforced top and bottom in the longitudinal and transverse directions by 20M (#6 bars). These were spaced at 230 mm in the transverse direction and 280 mm in the longitudinal direction.

The specimens were identified as *North*, *Center* or *South* depending on their location in the original bridge deck. Although the six specimens were geometrically identical, test coupons provided by the fabricator indicated that the yield strength of the A36 steel in the two bridge models were 310 and 370 MPa. Thus, three of the specimens had wide flange beams with a yield stress of 310 MPa and three others had a yield stress of 370 MPa. In the following, specimens are identified by their position in the original bridge deck and the yield stress that is used as a suffix, e.g. *North-310 MPa*, *Center-370 MPa*, etc.

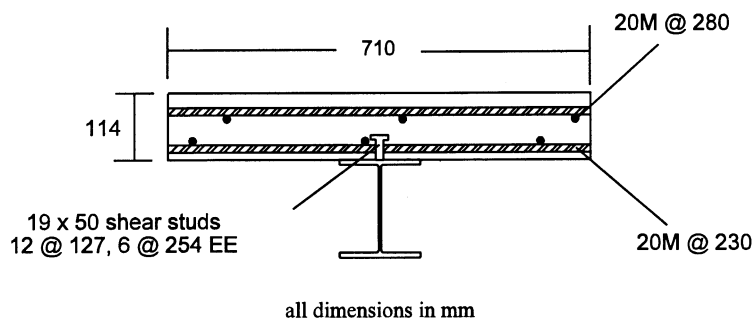


Fig. 2. Slab reinforcement details.

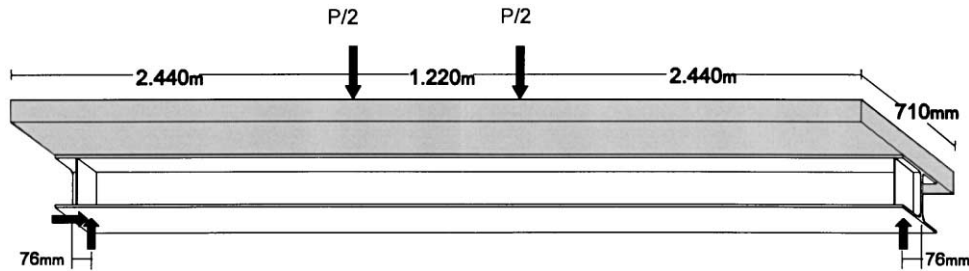


Fig. 3. Loading of simply supported steel composite bridge section.

## 4. Materials

### 4.1. Concrete

Normal weight concrete used in casting the bridge deck was supplied locally and was selected to satisfy Florida Department of Transportation's specifications for Class II concrete. Specialist help was used for pouring, screeding and finishing. The 28 day compressive strength of 43.6 MPa was 40% higher than the 31 MPa design strength. An additional 18% compressive strength gain was realized at the time of testing, which resulted in a compressive strength range of 48.5–52.6 MPa.

### 4.2. CFRP laminates

The CFRP laminates used were fabricated by Composite Horizons, Covina, CA using a simple hand layup process. Widely available T300 carbon fibers and an epoxy matrix were used. The laminate had a uni-directional elastic modulus of 114 GPa and an average unidirectional tensile strength of 1840 MPa. A linear stress-strain relationship gave an ultimate failure strain of 1.6%. Three of the six laminates were 2 mm thick, the remaining three were 5 mm thick. These thicknesses are considerably greater than those used for strengthening concrete or wood and were established from non-linear finite element analyses to ensure comparable strengthening [9]. The width of the laminate was 0.165 m to match the width of the flange of the W8 × 24 beam but its length was only 3.65 m, being limited by fabrication considerations of the CFRP laminate.

### 4.3. Adhesive

The two-part epoxy adhesive FR1272 was selected based on the recommendation of the CFRP laminate fabricator. It is relatively inexpensive and is widely available. However, it has a pot life of 10–15 min at 24°C making its field applicability limited.

## 5. Simulation of damage

In order to meet the objectives of this study, it was first necessary to simulate severe service distress in the

specimens. This was achieved by pre-loading each specimen (see Fig. 3 for schematic arrangement) past yield of the tension flange, a much higher stress level than the maximum stress of 55% of yield permitted by the AASHTO Bridge Specifications [1]. As a result of this loading, there was permanent deformation in each member prior to strengthening. A summary of the applied loads, deflections and maximum recorded steel strains for all the specimens appears in Table 1. The applied loads averaged 142 and 187 kN for the 310 and 370 MPa girders, respectively.

## 6. Repair of damaged specimens

### 6.1. Beam/CFRP combination

Since there were two different sets of yield strengths (310 and 370 MPa) and two different CFRP laminate thicknesses (2 and 5 mm), several beam/laminate combinations were considered. However, to obtain the most data, each steel type was strengthened using each of the two different CFRP laminate thicknesses.

Two of the three lower capacity 310 MPa steel composite beams were strengthened using thicker 5 mm CFRP laminate since this would provide a measure of the maximum strength gain that was attainable. The remaining specimen was strengthened using the 2 mm laminate. This meant that two of the three 370 MPa specimens were strengthened using 2 mm CFRP laminates (providing a lower bound on strength gain) and one by the 5 mm laminate.

### 6.2. Adequacy of adhesive

Because the epoxy adhesive is so much weaker than steel, non-linear finite element analyses were carried out to evaluate its adequacy under ultimate conditions. The results of the analyses indicated that the epoxy would fail at the ends of the CFRP laminate due to high peeling stresses. To prevent this type of failure, a three-piece steel clamp was designed so that it could be positioned precisely where needed to withstand the predicted peeling stresses. The clamps were attached to the bottom flange of the W8 × 24 section using two, 19 mm A490 bolts and two smaller 8 mm hardened steel bolts. Only two sets of clamps were needed since they could be re-used. Brief details on the analysis and

Table 1  
Summary of results

| Specimen ID<br>(CFRP thickness) | Mid-span values           |                    |                        |                   |                    |                         |                     | Plain section<br>ultimate capacity<br>(kN) | Strength gain<br>(%) | Comments                  |
|---------------------------------|---------------------------|--------------------|------------------------|-------------------|--------------------|-------------------------|---------------------|--------------------------------------------|----------------------|---------------------------|
|                                 | Simulating service damage |                    |                        | Repaired specimen |                    |                         |                     |                                            |                      |                           |
|                                 | Load<br>(kN)              | Deflection<br>(mm) | Tensile strain<br>(μϵ) | Ult. load<br>(kN) | Deflection<br>(mm) | Concrete strain<br>(μϵ) | CFRP strain<br>(μϵ) |                                            |                      |                           |
| North 310 (2-mm)                | 142                       | 40.90              | 1524                   | 237               | 102.9              | 2258                    | 6646                | 196                                        | 21.0                 | No failure bolts          |
| Center 370 (2-mm)               | 186                       | 44.96              | 1744                   | 271               | > 100.0            | 2267                    | 8164                | 249 <sup>a</sup>                           | 8.7                  | No failure no bolts       |
| South 370 (2-mm)                | 187                       | 44.96              | 1717                   | 272               | > 100.0            | 2284                    | 7811                | 249 <sup>a</sup>                           | 9.1                  | No failure no bolts       |
| Center 310 (5-mm)               | 142                       | 39.69              | 1239 <sup>b</sup>      | 298               | 96.0               | 1918                    | 5099                | 196                                        | 52.2                 | CFRP rupture along bolts  |
| South 310 (5-mm)                | 142                       | 30.23              | 1275 <sup>b</sup>      | 169               | 32.3               | 786                     | 1538                | 196                                        | − 13.7               | Adhesive failure no bolts |
| North 370 (5-mm)                | 187                       | 45.72              | 1695                   | 329               | 87.9               | 1778                    | 4859                | 249 <sup>a</sup>                           | 32.0                 | CFRP rupture along bolts  |

<sup>a</sup> Predicted value.

<sup>b</sup> indicates gage failure at specified strain (after value).

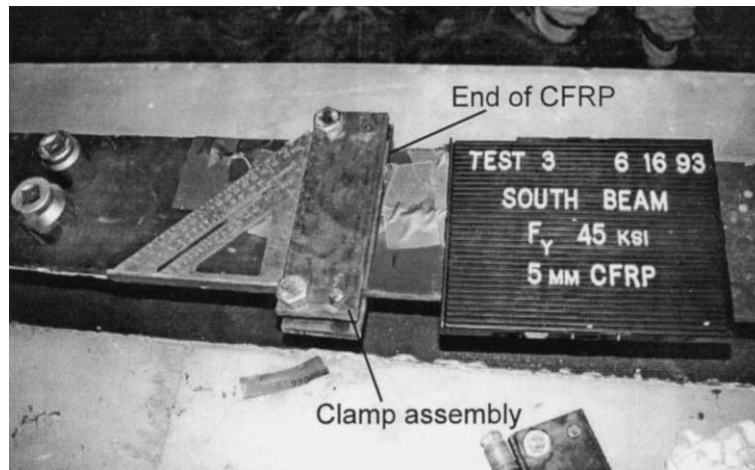


Fig. 4. Clamp assembly used to withstand peeling stresses.

design of the clamps may be found elsewhere [8]. Complete details of the study are presented in Ref. [6].

### 6.3. Bonding CFRP to steel

Surface preparation is important for ensuring a successful bond between CFRP laminate and steel. Since the steel girders used in the bridges were provided with a protective coat of paint, it was necessary to remove this paint. The specimens were turned over to provide access to the bottom flange where the CFRP was to be bonded and a portable sandblaster was used to obtain the necessary etched surface. The CFRP laminate was bonded to the steel surface using the epoxy adhesive in accordance with the manufacturer's recommendations. After the adhesive cured for 48 h, the specially designed clamps resisting peeling stresses were positioned over the ends of the laminate and secured in place using bolts as shown in Fig. 4.

## 7. Testing of repaired specimens

### 7.1. Instrumentation

The test specimens were most heavily instrumented at the quarter and mid-span locations. Strain gages were embedded inside the deck slab during casting and were also attached to the top concrete surface and to the underside of the steel flanges. Strain gages were also affixed to the CFRP laminate at mid-span and at 50 mm from each end of the 3.65 m long CFRP laminate, i.e. 1.27 m from the end of the beam. Additionally, gages were also attached to the bottom flange of the steel beam just beyond the CFRP laminate (1.14 m from the end of the beam). Comparison of strains in the steel beam and that in the CFRP laminate in close proximity allowed slip to be readily detected.

LVDTs were used to measure deflections at the mid-span and quarter-span locations. Load cells were used to measure applied loads and support reactions. All data were auto-

matically recorded by a SYSTEM 4000 data acquisition system.

### 7.2. Loading

The same four-point loading scheme used for simulating service damage (see Fig. 3) was used for testing the repaired specimens. The applied load was transmitted through two 0.71 m long stiffened spreader beams that were welded to a 1.22 m long stiffened cross beam. The length of the spreader beams was the same as the width of the concrete slab. This meant that the test specimens were subjected to two uniform, transverse line loads spaced 1.22 m apart.

The crossbeam was directly connected to the ram of a 45 ton hydraulic jack using a swivel head that was attached by custom-designed connectors. This ensured that the loads were transferred equally to the two spreader beams. Four 150 mm × 200 mm fabric pads were placed under both spreader beams to ensure uniform distribution.

### 7.3. Adhesive failure

The first two specimens tested were the *Center-* and *South-*370 MPa steel composite beams strengthened using 2 mm CFRP laminates. The test was stopped when no additional load could be applied (271 and 272 kN) and the deflection exceeded the 100 mm limit of the LVDT (see Table 1). There was no adhesive failure, as was predicted by non-linear analysis.

The third specimen tested was *South-*310 MPa steel composite beam strengthened using 5 mm CFRP laminate. This test ended suddenly with the separation of the CFRP laminate from the steel flange. The clean separation was believed to be adhesive failure due in part to a manufacturing defect that left a small depression on the bonded face of the 5 mm thick CFRP laminate.

To prevent a recurrence of this failure, bolts were used to augment the load transfer capacity of the epoxy adhesive. The configuration and arrangement of bolts was based on

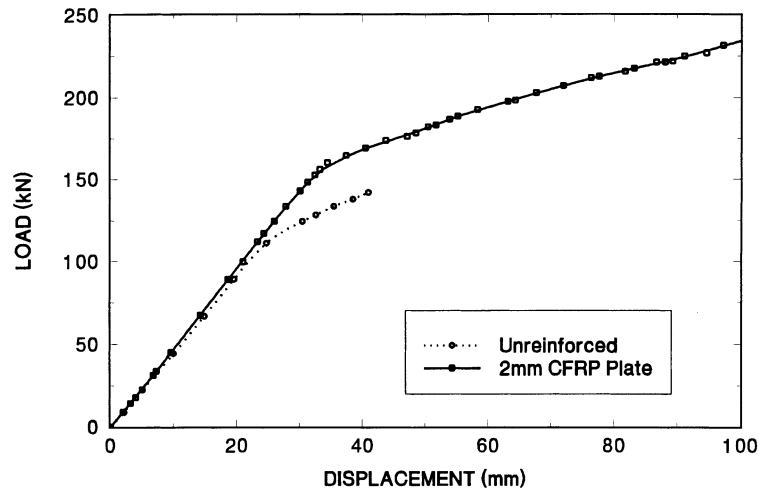


Fig. 5. Load-displacement *North*-310 MPa-2 mm (epoxy + bolted).

non-linear finite element analysis and consisted of two lines of three, 19 mm A490 bolts spaced at 41.9 cm, starting 8.9 cm from the edge of the laminate [6]. In all, a total of 12 steel bolts were used (six at each end) in the two remaining tests involving 5 mm thick CFRP laminates. Although no bolts were needed for the 2 mm CFRP laminate, they were provided nonetheless to assess the effect of load transfer in the presence of bolts. A pair of bolts was provided 8.9 cm from each end of the CFRP in the last specimen strengthened using 2 mm, (for details refer to Ref. [8]).

## 8. Results for repaired beams

### 8.1. Strengthening

Load vs displacement plots comparing the responses of the six repaired and plain specimens are presented in Figs. 5–10. The results for specimens repaired with 2 mm CFRP laminates appear in Figs. 5–7, and those for the 5 mm CFRP

laminates in Figs. 8–10. A summary of the maximum loads and mid-span deflections and strains in both concrete and the CFRP laminate is presented in Table 1. This table also contains information on the load applied to simulate damage referred to earlier.

Table 1 shows that with one exception *South*-310 MPa, increases in the ultimate strength were substantial. For example, *Center*-310 MPa realized an increase in strength from 142 to 298 kN, a strength gain of 110%. Inspection of Figs. 5–10 shows that the strengthening effect is largely confined to the post-yield region. This is not altogether surprising since the CFRP laminate has a lower elastic modulus –56% that of steel. This meant that prior to yielding, stresses in the composite laminate were lower than those in the bottom steel flange.

The premature local adhesive failure of *South*-310 MPa, enabled it to be re-tested to determine the ultimate capacity of the steel section (see Fig. 9). The results of this test were used to fine-tune the non-linear finite element model to predict the ultimate capacities of all the specimens that

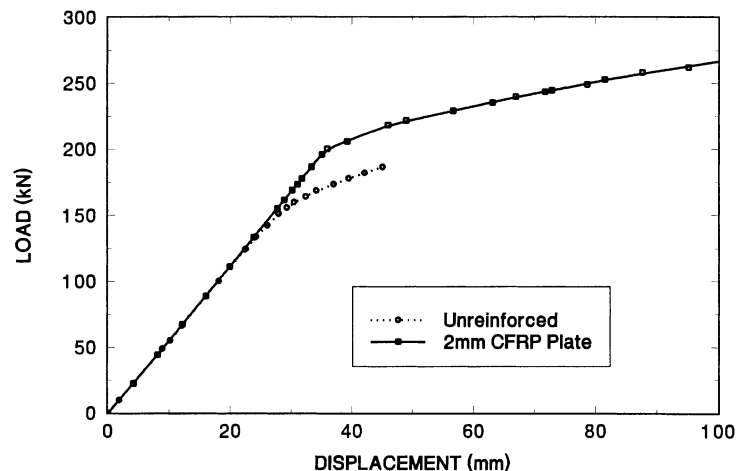


Fig. 6. Load displacement *Center*-370 MPa-2 mm (epoxy only).

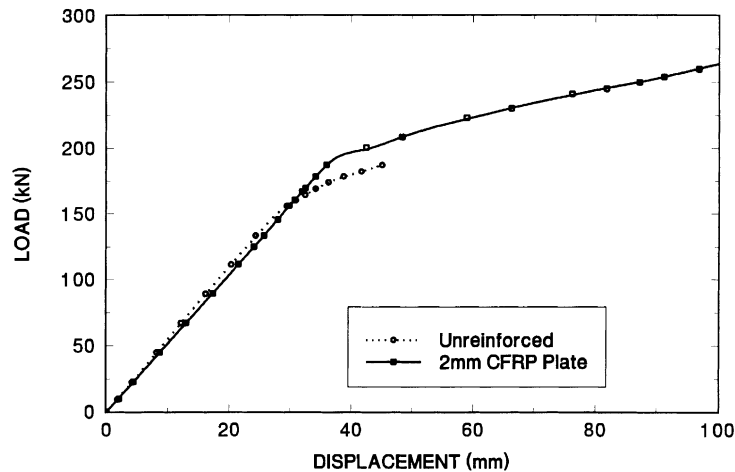


Fig. 7. Load displacement *South-370 MPa-2 mm* (epoxy only).

were tested. The predicted value was found to be 249 kN for the 370 MPa girders. Using the measured and the predicted failure loads, the *absolute* increases in ultimate strength of the strengthened steel beams were found to range between 9 and 52% (Table 1). The smaller increases correspond to the 2 mm laminate and the larger increases to the 5 mm ones.

In addition to increases in ultimate load, the elastic region of the plain section was considerably extended in the repaired specimen. The increases ranged from about 20% for *South-370 MPa/2 mm* (Fig. 7), where the elastic range was extended from 156 to 187 kN, to over 67% in the case of *North-370 MPa/5 mm* (Fig. 10). This increase is particularly encouraging since it suggests that service loads of repaired specimens could be increased using this technique.

## 8.2. Stiffening

The load vs deflection response of the strengthened and unstrengthened beams also provide a measure of the stiffen-

ing effect in the elastic region. This is an important consideration particularly for bridge application where live load deflection limits are very stringent.

Figs. 5–10 show that there is very little stiffening in the elastic region except for specimens strengthened with the 5 mm CFRP laminates (Figs. 8–10). Even for this case, there was practically no stiffening for *Center-310 MPa* specimen (Fig. 8) due to cracking in the slab from the initial loading. For specimens strengthened using 2 mm laminates there is even a reduction in stiffness in the elastic region (Fig. 7). This is probably because the concrete damage in this beam could not be fully offset by repairs and the stiffness lost as a result was not regained by bonding the CFRP laminate.

The lack of increase of stiffness in the elastic range is due to CFRP's smaller elastic modulus compared with steel. The increase in stiffness resulting by bonding a CFRP laminate to the steel surface farthest from the neutral axis is partially offset by this lower modulus. However, with thicker

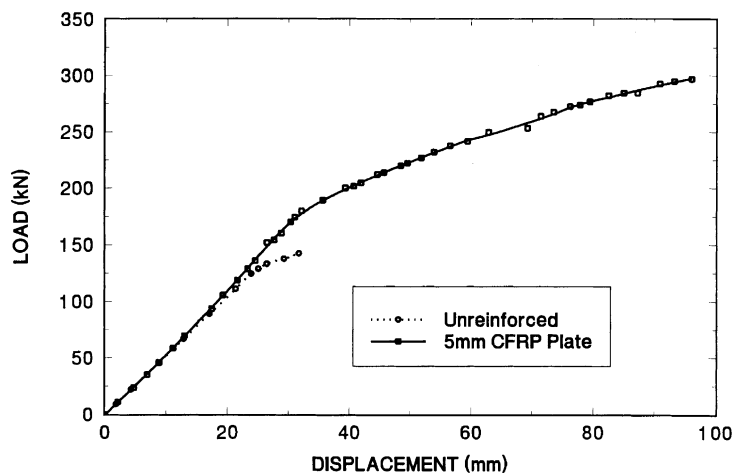


Fig. 8. Load displacement *Center-310 MPa-5 mm* (epoxy + bolted).

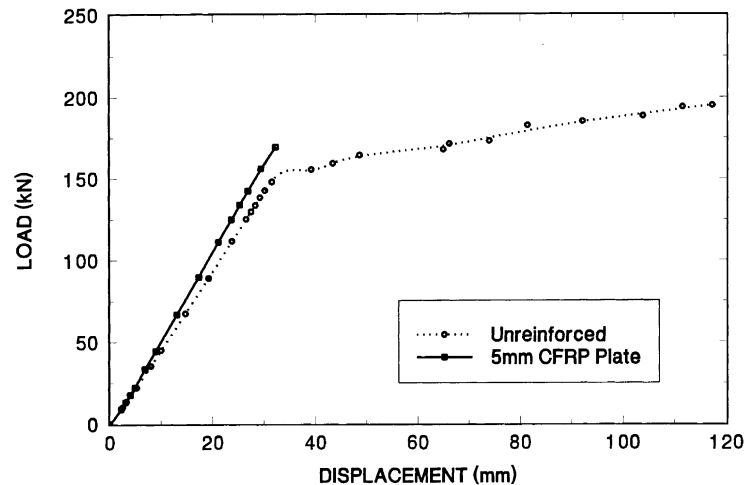


Fig. 9. Load displacement *South*-310 MPa-5 mm (epoxy only).

laminates, increases are larger provided there is no adhesive failure (Fig. 10). If local adhesive failures occur, strengthening cannot be expected (e.g. Fig. 9).

### 8.3. Failure mode

The load deflection plots in Figs. 5–10 indicate that, in general, there was a considerable amount of deflection when the maximum load was reached (Fig. 11). For the beams strengthened using 2 mm CFRP the test was stopped when no additional load could be applied. As such, there was no failure of the CFRP laminate or its bond with the steel surface. With the beams strengthened using 5 mm CFRP, however, failure occurred due to shearing off the CFRP laminate by the bolts (see Fig. 12).

### 8.4. Bolted vs bonded

In order to assess the relative performance of bonded and bolted specimens, the strain variations with depth at the critical mid-span locations for different cases are shown in

Figs. 13–16. The results for the 2 mm CFRP laminates are shown in Figs. 13 and 14 while those for the 5 mm laminates in Figs. 15 and 16.

Inspection of Figs. 13 and 14 showing the strain variation for the 2 mm CFRP laminate indicate their similarity. This is not surprising since there was no separation of the 2 mm laminates, whether bonded or bolted. In each of these cases, the load was transferred primarily by the adhesive.

The comparison of strain variation along the depth of the specimen strengthened with 5 mm CFRP laminate shown in Figs. 15 and 16 indicate obvious dissimilarities. For the bonded specimen (Fig. 15) there is load transfer to the laminate by the epoxy until adhesive failure. The bolted specimen (Fig. 16) transfers load in the same manner until the CFRP delaminates from the steel at about the same load. The loads hitherto being carried by the epoxy were now transferred to the bolts with commensurate strain relaxation in the CFRP laminate. Failure finally occurred when excessive strain (and therefore deformation) around the bolts led to rupture of the CFRP laminate (Fig. 12). This

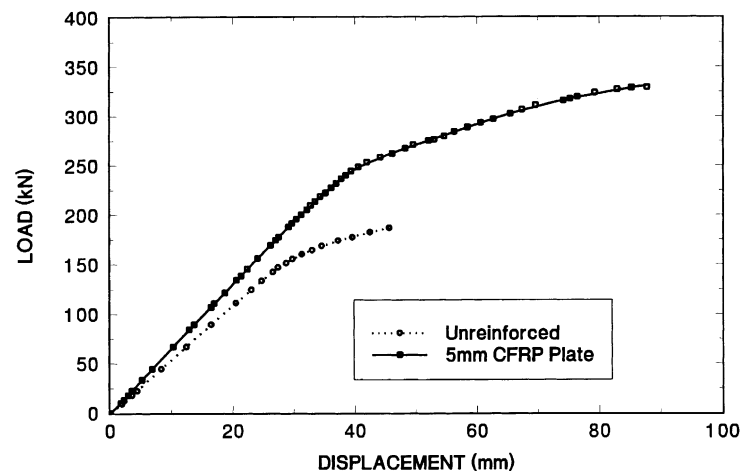


Fig. 10. Load displacement *North*-310 Mpa-5 mm (epoxy + bolted).



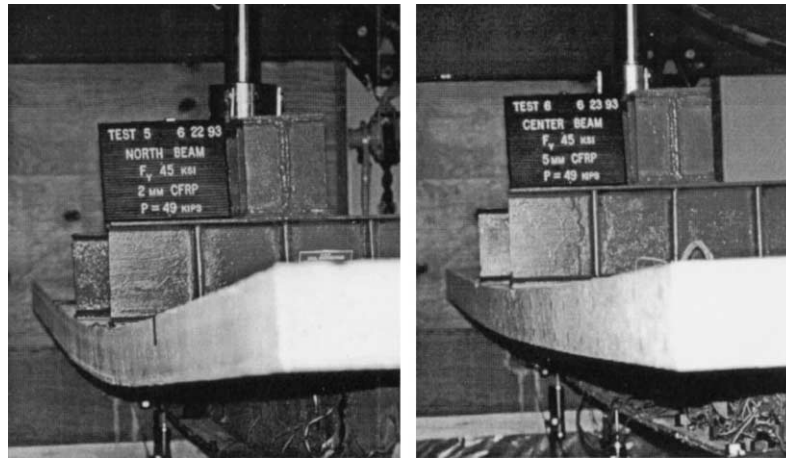


Fig. 11. Deflected shape of repaired 310 MPa beams at 218 kN.

problem may be avoided by using a greater number of bolts, thereby limiting their maximum strain and by using CFRP laminates manufactured with bolt holes and multi-directional carbon fiber reinforcement around such holes.

## 9. Discussion

The results from the study indicate clearly that it is possible to strengthen steel composite bridge sections using CFRP laminates. Because of the high strength and stiffness of steel the operation is somewhat more complicated than that used for concrete or wood. In this study, a combination of adhesive, fasteners, and clamps were used.

Because T300 carbon fibers were used for cost reasons, strengthening was more pronounced in the post-yield region than in the elastic region. However, if CFRP laminates with higher stiffness are used, an increase in the elastic region may be expected. With the prices of carbon fiber dropping, this may be realizable earlier than previously thought possible.

Since the epoxy adhesive is very weak in comparison with steel, bolting was required especially to assist in the load transfer for the thicker laminates. In configurations that

dictate transferring shear using bolts, the use of adhesives would still be recommended to help distribute the load and provide transfer of shear until the bolts become fully loaded in bearing. In this case, the use of adhesives served to secure the CFRP laminate while the bolt holes were drilled. It would be preferable to use a larger quantity of smaller bolts to minimize the magnitude of localized stress concentrations at the bolts and to reduce the tendency for transverse CFRP laminate fractures. The magnitude of the localized stress concentrations may also be reduced by implementing high quality drilling techniques, and by utilizing CFRP laminates which have more plies with varying fiber orientation in the bolt hole regions, or by using CFRP laminates specially manufactured to include reinforced holes for bolting. Installation would be facilitated by such manufactured holes or by pre-drilling the abrasive, drill-resistant CFRP laminates prior to transport to a field site.

Although bolts and clamps were successfully designed, sized, and positioned to aid in shear transfer and to resist peeling stresses at the laminate ends, the use of bolts somewhat negates the time efficiency of CFRP repairs. To ensure effective repairs the bond interface of the CFRP must be free from defects at the onset to maintain uniform adhesive thickness and reduce incidence of local adhesive failures.

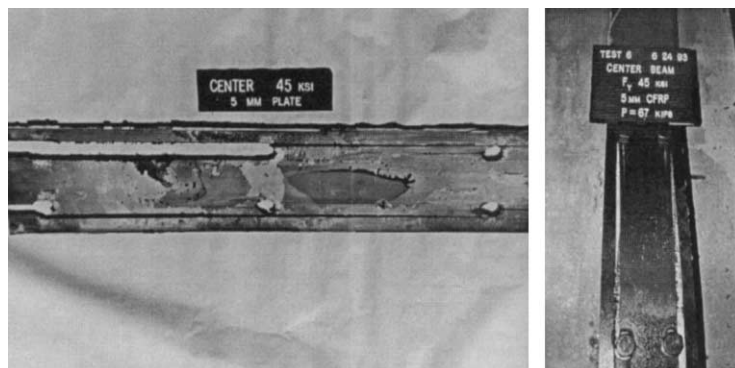


Fig. 12. Rupture of 5 mm CFRP plate (Center-310 Mpa).

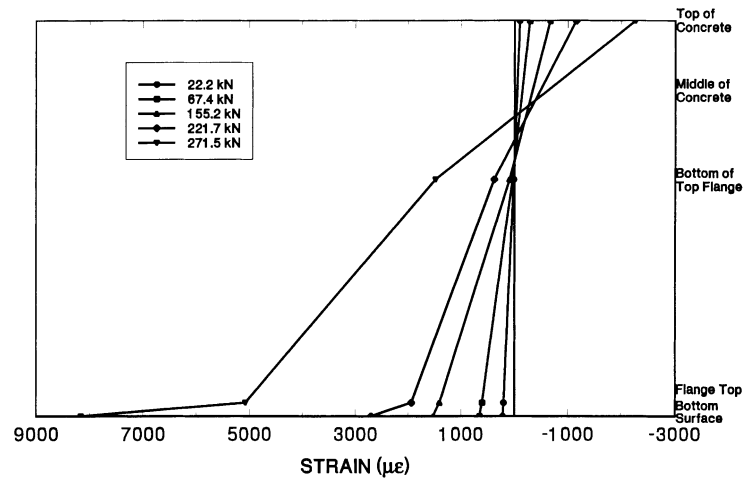


Fig. 13. Strain variation at mid-span *Center*-370 MPa-2 mm (epoxy).

Additionally, tapered laminates in both width and thickness accompanied by a scheme of adhered tapering with an adhesive fillet can reduce the stress concentrations leading to high peeling stresses.

### 10. Non-linear finite element analysis

The non-linear finite element analysis program PCFRAME [5] was used to predict the response of the six steel composite bridge sections tested. The concrete slab was discretized into 20 layers and the steel reinforcement by two. The wide flange section was modeled as eight steel layers with the CFRP laminate and the adhesive represented by single layers (see Fig. 17). Along the length, each beam was discretized using 41 members.

Figs. 18–20 show comparisons of predictions from the non-linear finite element analysis with test results. The strain variation in the CFRP laminate and concrete strain

are shown in these figures. Additional results, including a complete set of comparisons for load vs deflection and steel strains for the unreinforced and strengthened beams tested, may be found elsewhere [2].

Inspection of Figs. 18–20 show excellent correlation between the experimentally measured strains and predictions from the finite element analysis. This is especially true with the 2 mm laminate. The correlation is somewhat poorer for the 5 mm laminate at higher loads. This is not altogether surprising since some of the load taken by the bolts could not be modeled. Use of bolts in the 2 mm specimens proved to have no effect as the adhesive sufficiently transferred the shear forces without causing bearing in the bolts.

Agreement for load vs displacement was not as close with the finite element analysis predicting a stiffer post-yield response. This was because the effect of pre-damage was only approximated in the analysis. The overall agreement between experiment and finite element analysis was generally quite satisfactory.

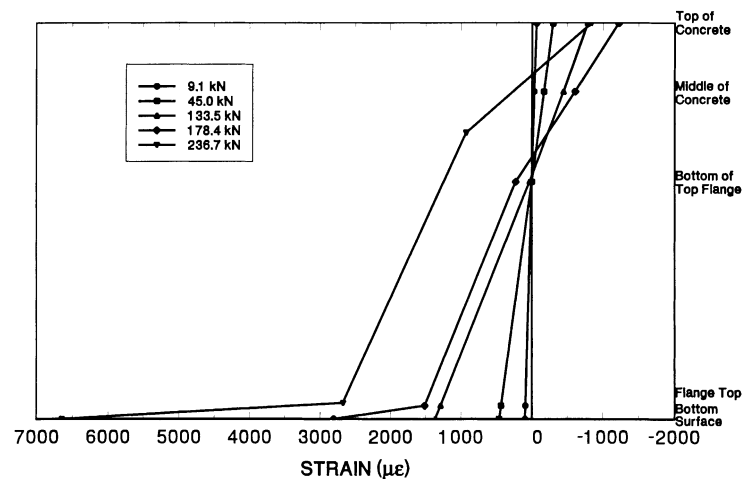


Fig. 14. Strain variation at mid-span *North*-310 MPa-2 mm (epoxy + bolted).

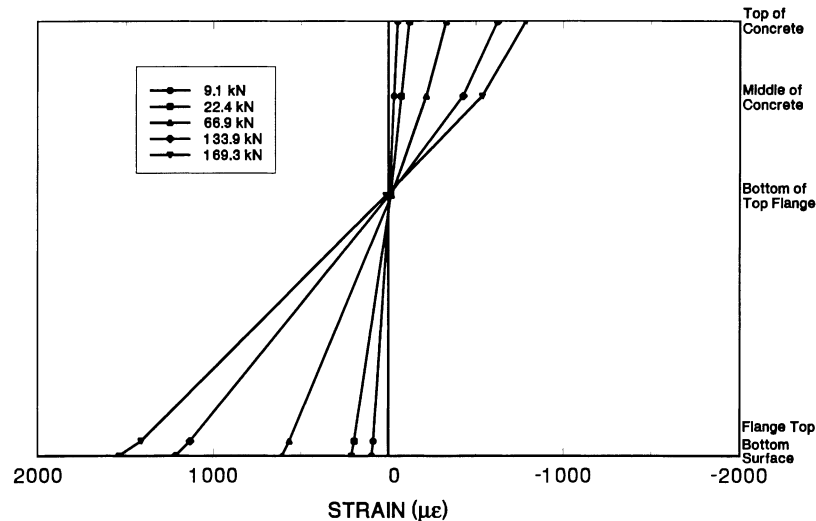


Fig. 15. Strain variation at mid-span *South*-310 MPa-5 mm (epoxy only).

## 11. Conclusions

On the basis of the experimental and numerical studies presented, the following conclusions may be drawn:

1. Steel composite members may be strengthened by bonding CFRP laminates. However, much thicker laminates are needed to achieve comparable strength and stiffness gains typically obtained in strengthened concrete or wood members. Additionally, the capacity of the epoxy adhesive to transfer loads to the CFRP laminates may need to be augmented, e.g. by clamps or fasteners, to ensure load transfer past yielding of steel.
2. Test results showed significant increases in ultimate capacity of steel composite bridge members strengthened by CFRP laminates. Strength gains for the bridge sections containing 310 MPa wide flange shapes were 21 and 52% for the 2 and 5 mm CFRP laminates, respec-

tively. Corresponding increases for the bridge sections containing 370 MPa wide flange shapes were 9 and 32%, respectively.

3. Increases in the elastic stiffness were relatively modest (see Figs. 5–10) particularly when thinner 2 mm CFRP laminates were used. However, increases in the elastic region of strengthened members were substantial indicating the potential for increases in service loading.
4. The failure mode of the strengthened sections was generally ductile and accompanied by considerable deformation (see Fig. 11). However, where only epoxy adhesive was used for bonding a 5 mm CFRP laminate, adhesive failure led to separation of the CFRP laminate and sudden failure (*South*-310 MPa). No similar separation was observed with 2 mm CFRP laminates bonded using epoxy alone in that smaller tensile forces developed in the thinner CFRP and the associated shear stresses never exceeded the epoxy's shear strength.

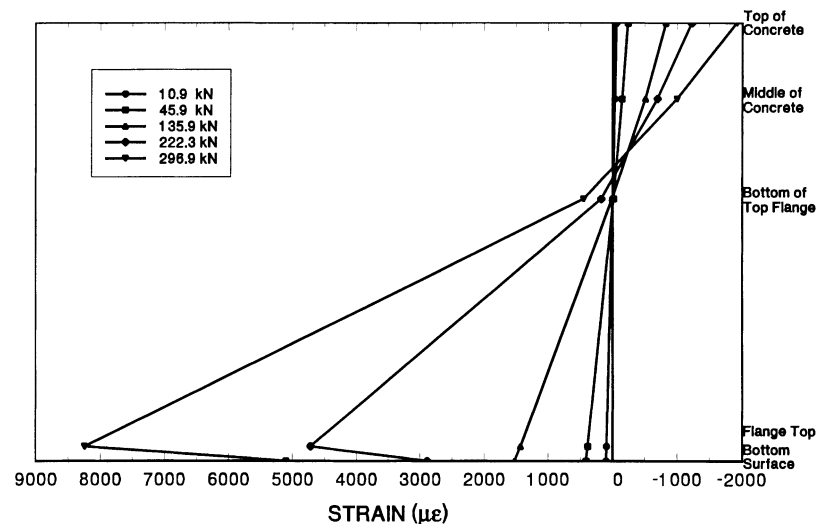


Fig. 16. Strain variation at mid-span *Center*-310 MPa-5 mm (epoxy + bolted).

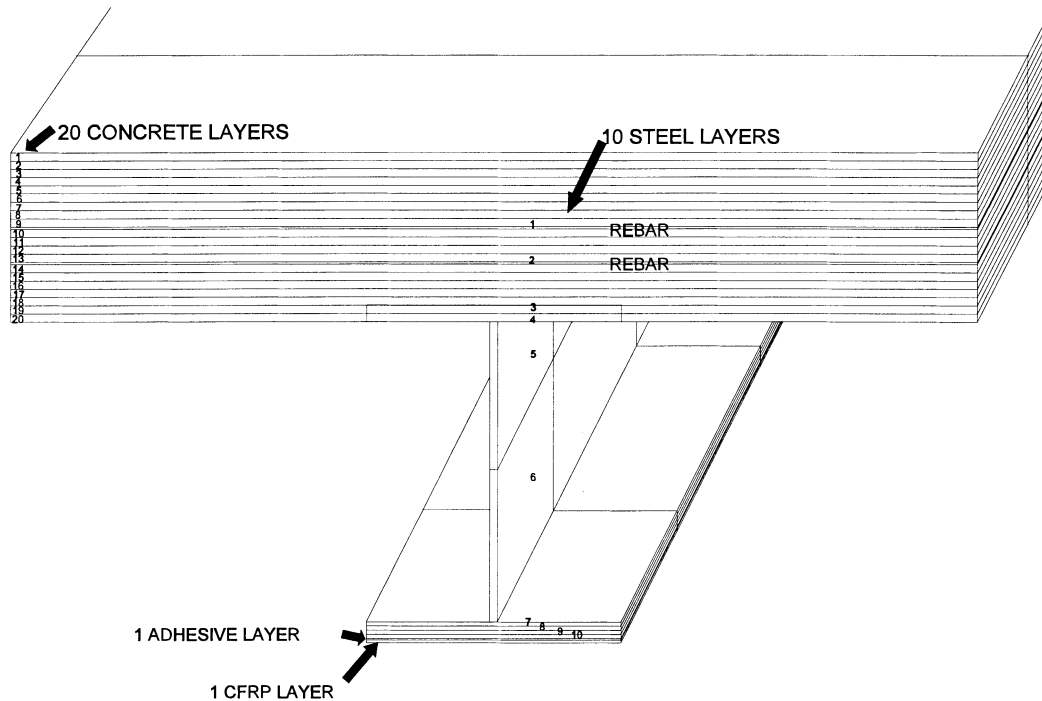


Fig. 17. Discretization of cross-section for finite element analysis.

5. The CFRP laminate must be adequately anchored to the steel flange to prevent peeling failure at the ends. In this study, a re-usable clamp system (see Fig. 4) was developed from non-linear finite element analysis that proved to be simple, efficient and inexpensive. These peeling forces could be minimized by sculpting the ends of the laminates and adhesive to reduce stress concentrations in this region. However, this would probably require ordering the laminates to case-specific lengths and would reduce field installation flexibility.
6. The response of the strengthened specimens was

predicted with reasonable accuracy by non-linear finite element analyses.

The study demonstrated the feasibility of using CFRP laminates for strengthening steel composite bridge members. If the intent of the installation is to increase the service load capacity and not to simply restore lost capacity due to damage, both the slab and shear connectors attaching it to the upper steel flange must have the reserve capacity needed for supporting the additional compressive and shear forces.

While clamps and fasteners were both used in the study,

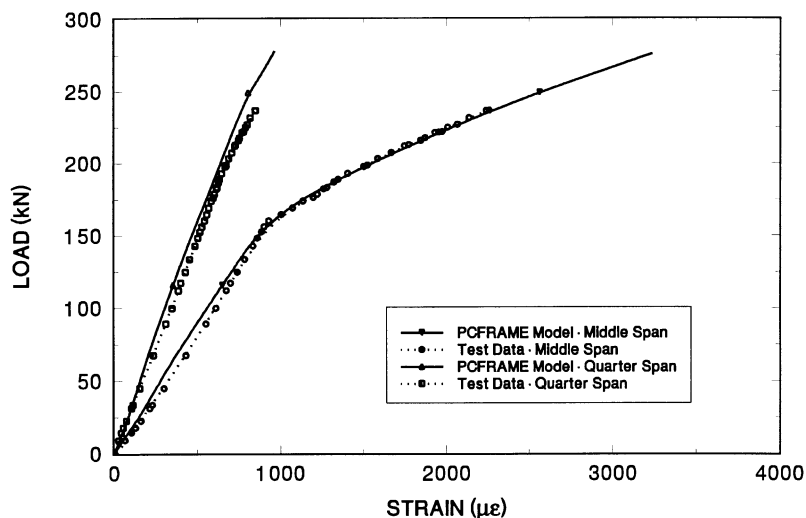


Fig. 18. Comparisons: load vs concrete strain (*North*-310 Mpa-2 mm).

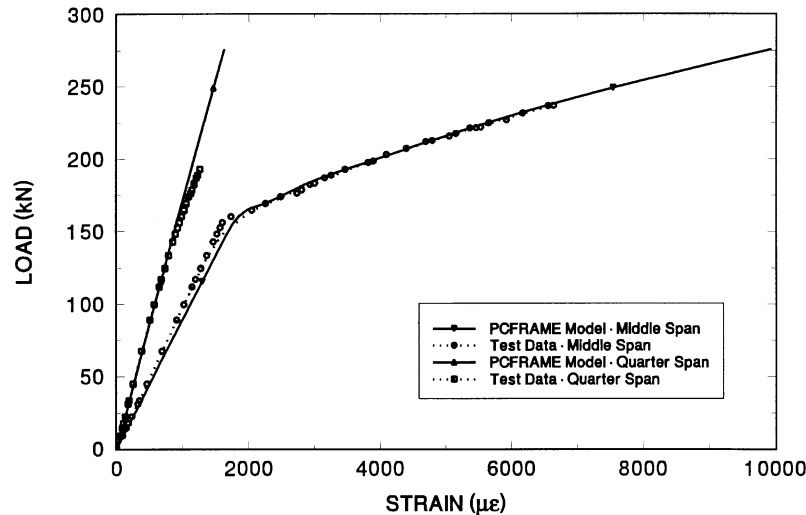


Fig. 19. Comparisons: load vs CFRP strain (*North-310 Mpa-2 mm*).

ideally clamps alone will be more desirable for field installation as it eliminates the need for heavy magnetic drills that will otherwise be needed to secure the CFRP laminates. This will greatly facilitate the speed with which repairs can be carried out particularly in congested urban areas — a very important consideration for using the light-weight CFRP laminates in the first place.

The effects of moisture and humidity on the long-term performance of the adhesive must be evaluated before field applications. However, this issue was not included within the scope of this study. In addition, appropriate measures developed by the aerospace industry should be used to prevent possible galvanic action due to the direct contact of the CFRP laminate with steel in the presence of moisture.

### Acknowledgements

The investigations reported were carried out with the financial support of the Florida Department of Transportation and the US Department of Transportation. The authors gratefully acknowledge the contribution of Dr Mohsen Shahawy, Director, Structural Research Center, Florida Department of Transportation and Mr Ken Spillett, formerly a graduate student in the Department of Civil and Environmental Engineering, University of South Florida, currently a Professional Associate, Parsons Brinckerhoff Construction Services, Tampa, FL. They also wish to express their sincerest appreciation to Composite Horizons, Covina, CA for their invaluable advice and assistance. However, the opinions,

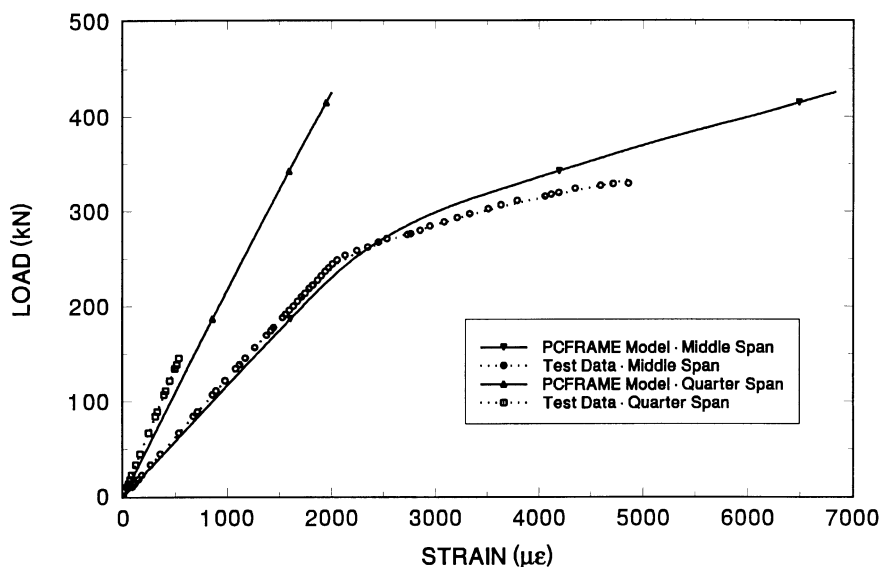


Fig. 20. Comparisons: load vs CFRP strain (*North-370 Mpa-2 mm*).

findings and conclusions expressed in this publication are those of the writers and not necessarily those of the Florida or US Department of Transportation.

## References

- [1] American Association of State Highway and Transportation Officials. Standard specifications for highway bridges. Washington, DC., 15th ed., 1994.
- [2] Liby L. Steel composite bridge sections strengthened by carbon fiber reinforced plastic laminates, MSCE Thesis submitted to Dept. of Civil Engng and Mechanics, University of South Florida, Tampa, FL, November, 1993. 353 p.
- [3] Mertz DR, Gillespie JW. Rehabilitation of steel bridge members through the application of advanced composites. Final Report, NCHRP-93-ID01, 1996.
- [4] Mosallam AS, Chakrabarti PR. Making the connection, Civil Engineering, ASCE, April, 1999. p. 56–9.
- [5] PCFRAME. Computer program for non-linear analysis of reinforced and prestressed concrete frames. Developed by Y. Kang, Ph.D dissertation, University of California, Berkeley, 1977.
- [6] Sen R, Liby L. Repair of steel composite sections using CFRP laminates. Final report submitted to Florida and US Department of Transportation, August, National Technical Information Service, VA, 1994.
- [7] Sen R, Spillett K. Restraint effect of bridge bearings. Final Report submitted to Florida and US Department of Transportation, January, National Technical Information Service, VA, 1994.
- [8] Sen R, Liby L, Spillett K, Mullins G. Strengthening steel composite bridge members using CFRP laminates. In: Taerwe L, editor. Non-metallic (FRP) reinforcement for concrete structures. London, England: E and FN Spon, 1995. p. 551–8.
- [9] Wang Y. Bridge strengthening using advanced composites, MSCE thesis submitted to the Department of Civil Engineering and Mechanics, University of South Florida, Tampa, FL, December, 1992.
- [10] Meier U, Deuring M, Meier H, Schwegler G. Strengthening of structures with CFRP laminates: research and applications in Switzerland. In: Neale KW, Labossiere P, editors. Advanced Composite Materials in Bridges and Structures. CSCE, Montreal, P.Q., 1992. p. 243–51.
- [11] Rostasy FS, Hankers C, Ranisch E-H. Strengthening of R/C and P/C structures with bonded FRP plates. In: Neale KW, Labossiere P, editors. Advanced Composite Materials in Bridges and Structures. CSCE, Montreal, P.Q., 1992. p. 253–63.
- [12] Uemura M, Aoyama K, Kliger H. A new reinforcing fiber sheet material and its application in the repair of concrete structures. In: 39th International SAMPE Symposium, April, Vol. 39, 1994. p. 347–54.
- [13] Shahawy M. Strengthening highway bridges with carbon fiber materials. In: 74th Annual Meeting of the Transportation Research Board, Washington, D.C., January, 1995.